

# Shervin Zare

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## EDUCATION

### Wilfrid Laurier University

*Bachelor of Science in Data Science*

Apr 2027

Waterloo, ON

- Relevant Coursework: Data Analytics, Databases, Data Visualization, Statistics, Software Engineering

## EXPERIENCE

### Educational Programs Developer

Sep. 2025 – Mar. 2026

*Laurier Analytics Society*

Waterloo, ON

- Developed educational content and video modules that simplified complex data and technology concepts, helping students deepen their understanding and stay engaged throughout the learning process.
- Delivered hands-on workshops in analytics, machine learning, and programming, giving students practical experience with real-world tools and improving technical confidence
- Organized major networking events with speakers from companies like Google and Sun Life to help students build meaningful industry connections.

### Data Analyst

May 2025 – Aug 2025

*Union Dome*

Toronto, ON

- Analyzed union member and platform usage data to identify engagement trends, supporting product improvements and enhancing overall platform performance.
- Developed operational reporting workflows using Excel, including structured tables, pivot tables, and scheduled updates, to deliver consistent, reproducible reports for product and client teams.
- Partnered with product and client teams to answer data-driven questions and provide insights that informed feature development and customer communication strategies.

## PROJECTS

### Customer Conversion Predictor 🧠 | *pandas, scikit-learn, XGBoost, Feature Engineering, Model Evaluation, seaborn*

- Developed a deployable XGBoost model for predicting term deposit subscriptions, achieving 82.8% accuracy and 83.7% recall, and packaged the final trained model within an end-to-end pipeline to facilitate immediate, data-driven targeting of high-probability customers and optimize marketing ROI.
- Implemented advanced feature engineering, pre-processing, and stratified sampling on imbalanced data, generating actionable insights through precision-recall analysis, ROC curves, and confusion matrices to improve conversion efficiency

### NetTrace 🧠 | *NetworkX, PyTorch Geometric, GraphSAGE, Scikit-learn, PowerPoint*

- Developed a graph-based fraud detection system leveraging a 2-layer GraphSAGE model, Isolation Forest anomaly detection, and community-based signals to identify fraudulent accounts in highly imbalanced financial data, achieving 81.1% precision
- Built and presented an end-to-end fraud detection solution at the Laurier Analytics Society Data Minds Hackathon, translating model insights into a PowerPoint presentation to communicate high-risk accounts and support data-driven fraud investigation.

### RAG-Powered Chatbot 🧠 | *LangChain, Streamlit, HuggingFace Embeddings, Vector Database, GPT-5.4 mini*

- Engineered a retrieval-augmented generation (RAG) chatbot using Streamlit and LangChain, integrating HuggingFace embeddings and an in-memory vector store to enable fast, context-aware document search and question answering.
- Enhanced user interaction with dynamic input modes (URL and file upload), automated text processing, and context visualization, providing actionable insights from unstructured documents.

### YUTH 🧠 | *Next.js, TypeScript, Supabase, OpenAI API, TailwindCSS*

- Built an agentic AI financial assistant that analyzes user profiles and contextual data to generate personalized recommendations for relevant government programs, benefits, and next-step actions, delivering real-time, data-driven financial guidance and recognized as a top 8 project out of 208 teams at Hack Canada.

## SKILLS

**Languages:** Python, C/C++, SQL, R, Java, Next.js, TypeScript

**Libraries:** pandas, NumPy, scikit-learn, LangChain, XGBoost, PyTorch, PyTorch Geometric, transformers, BeautifulSoup, requests, seaborn, Matplotlib

**Tools:** Git, VS Code, Jupyter, Excel, PowerPoint, Tableau, Power BI, Supabase, Streamlit, Plotly